# Modelling motion with graphs

**STUDENT WORKSHEET**

Name: _____ Class: _____ Date: _____

## Multiple choice

1. Which quantity is the gradient of an x–t graph? A acceleration B velocity C distance D force

2. Which quantity is the gradient of a v–t graph? A velocity B position C acceleration D speed

3. The gradient of an x–t graph is −3 m/s. The object is: A stationary B moving in −x C moving in +x D moving in a circle

4. A horizontal v–t line at +5 m/s describes: A rest B increasing speed C constant positive velocity D negative acceleration

5. A curved position–time graph becoming steeper with time shows: A constant speed B increasing speed (for positive slope) C zero velocity D necessarily negative acceleration

## Plot and explain

6. Plot position x (m) against time t (s), for t: 0,1,2,3,4,5 and x: 0,3,6,9,12,15. Label axes with units, draw best-fit line and calculate gradient. What is the velocity?

7. A car moves from x = +8 m at t = 0 s to x = −4 m at t = 4 s in a straight line at constant velocity. Sketch an x–t graph and state its gradient.

8. Sketch x–t graphs for (a) stationary at x = +4 m (b) moving forwards at constant velocity (c) reversing direction after moving away. Write a sentence for each.

9. Plot a v–t graph from the data: t (s): 0,1,2,3,4; v (m/s): 8,6,4,2,0. Calculate its gradient, with units.

10. A v–t graph is a horizontal line at −2 m/s. Describe velocity and acceleration. Sketch a possible x–t graph.

11. Consider a curved x–t graph that rises ever more steeply. Explain how you would determine the instantaneous velocity at t = 3 s.

12. Extension: A car has v = +4 m/s for 3 s, then v = −2 m/s for 2 s. Sketch its v–t graph and find net displacement by the signed areas.

## Motion-sensor challenge

13. For each graph demonstrated in class, describe how you would move relative to the sensor to match it: flat x–t, upward sloping x–t, downward sloping x–t, and increasingly steep upward x–t.